SuperStore Sales Dashboard

1. Business problem

Sales managers need a consolidated view of revenue, profitability, orders, delivery performance, geography and product mix. Reviewing individual order lines makes it difficult to compare periods and regions or identify low-margin segments. This project uses Power BI to turn the supplied Excel sales records into an interactive decision-support dashboard.

2. Objectives

- Monitor sales, profit, order volume and average shipping delay.
- Compare year-over-year monthly trends.
- Analyze sales by region, payment mode, segment, ship mode, category and subcategory.
- Explore state-level sales and profit using a map.
- Apply a region slicer consistently across compatible visuals.

3. Source inventory

Excel: 1 sheet, 5,901 order-line rows and 23 columns. Power BI: 17 report visual containers, including 4 cards, 3 doughnut charts, 3 clustered bar charts, 2 stacked area charts, one map and one slicer. Dashboard screenshot is a separate visual reference.

4. Independently calculated source measures

Order-line records: 5,901; distinct Order IDs: 3,003; sales: 1,565,804.32; profit: 175,262.11; quantity: 22,317; profit margin: 11.19%. Source-year distribution: 2019: 746 rows, 2020: 939 rows. Date-order anomalies (Ship Date earlier than Order Date): 0. Dashboard cards are rounded and may use different measures/filter context; do not assume they exactly match these source totals.

5. Data terminology and field dictionary

Term / field	Definition / use
Row ID+O6G3A1:R6	Row-level identifier; header appears corrupted or concatenated in the supplied file.
Order ID	Order identifier; multiple product rows can share one order.
Order Date	Date of order placement; Excel serial date.
Ship Date	Recorded shipping date; verify against order date.
Ship Mode	Shipping service class.
Customer ID	Customer identifier; pseudonymize before publication.
Customer Name	Customer name; remove from public sample.
Segment	Consumer, Corporate or Home Office customer group.
Country	Country of sale.

City	Customer city.
State	US state used for mapping.
Region	Central, East, South or West.
Product ID	Product identifier.
Category	High-level product family.
Sub-Category	Detailed product group.
Product Name	Descriptive product label.
Sales	Sales amount on an order line; currency not documented.
Quantity	Units on an order line.
Profit	Reported profit on an order line; source calculation not specified.
Returns	Return flag/status; contains #N/A and requires normalization.
Payment Mode	Payment channel, including COD, Online and Cards.
ind1 / ind2	Additional fields; source semantics not documented.

6. KPI definitions

Term / field	Definition / use
Total sales	SUM(Sales); respects active report filters.
Total profit	SUM(Profit).
Profit margin	DIVIDE([Total Profit],[Total Sales]).
Orders	DISTINCTCOUNT(Order ID); differs from row count.
Quantity sold	SUM(Quantity).
Average delivery days	AVERAGEX(valid order lines, Ship Date − Order Date); order-level version requires distinct-order aggregation and a stated rule.
Monthly sales	Total sales grouped by Order Date year and month.
Monthly profit	Total profit grouped by Order Date year and month.
Sales by state/category/etc.	Total sales grouped by the named dimension.

7. Quality and interpretation limitations

- Excel has repeated order IDs because one order may contain multiple order lines.
- Check inconsistent dates and invalid/blank return flags before reporting delivery or returns.
- The first column header is malformed; ind1 and ind2 lack definitions.
- Do not describe reported Profit as net earnings without source accounting definitions.
- Check whether the Power BI model applies filters or transformations that change the displayed totals.
- Customer names are present in the source; sanitize data before public GitHub upload.

- The Power BI report is not an Excel macro workbook; VBA does not run inside Power BI.

8. Tools and deliverables

Microsoft Excel (data source), Power BI Desktop, Power Query, DAX, Power BI visuals and slicers. Deliverables: README.md, this documentation, a complete rebuild tutorial, and the original PBIX/Excel files if publication rights permit.